

3(a)	any five from: max four from mp1–mp6: 1 (sickle cell anaemia is) homozygous recessive ; 2 ref. to beta / β, globin / polypeptide or <i>HBB</i> (allele / gene) ; 3 base / nucleotide, substitution or mis-sense mutation or change in, triplet / codon or GAG to GTG / A to T ; 4 causes, change in amino acid (sequence) / glutamic acid to valine ; 5 valine is non-polar ; 6 causes change in, 3D shape / tertiary structure / quaternary structure / globular structure (of haemoglobin) ; 7 haemoglobin has lower affinity for oxygen ; 8 (in red blood cells) haemoglobin, less soluble / sticks together / forms fibres ; 9 (red blood) cells become sickle shaped ;	5
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Question	Answer	Marks
3(b)(i)	decrease (in ADH) (accept no ADH secreted) and reduce water reabsorption (in the collecting duct) or decrease (in ADH) and increased volume of / more dilute, urine (formed) ;	1
3(b)(ii)	<i>idea that</i> fluid / water, is forced (from blood) into alveoli ; R diffusion / osmosis	1
3(c)	<i>any four from:</i> <i>describe:</i> 1 having kidney disease increases the risk of TACO ; ora 2 AKI has highest risk (of TACO) ; 3 data quote to support mp1 (comparison) ; <i>reasons:</i> 4 kidney disease reduces ability to remove excess water ; 5 more water retained increases blood volume ; 6 AVP ; e.g. CKD + D less risk as dialysis helps to restore water volume	4

Question	Answer	Marks
3(d)	any two from: 1 fewer sodium (ions) reabsorbed / more sodium (ions) in filtrate ; 2 decreases water potential of filtrate or increases water potential of the medulla ; 3 less water reabsorbed, by osmosis / down water potential gradient ; 4 increases volume of urine ; <i>plus:</i> 5 so blood volume (from transfusion) is reduced ;	3